An unconstrained Euclidean parametrisation of phylogenetic tree space

Xavier Didelot[1,2,*]

[1] School of Life Sciences, University of Warwick, Coventry, United Kingdom

[2] Department of Statistics, University of Warwick, Coventry, United Kingdom

[*] Corresponding author. Tel: 0044 (0)2476 572827. Email: xavier.didelot@warwick.ac.uk

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1 Introduction

Phylogenetic trees are usually represented using two components: the topology of the tree which is discrete and the branch lengths of the tree which are continuous (Paradis 2006). This mixed nature makes it difficult to explore the tree space, using for example moves such as nearest-neighbor interchange (NNI) or subtree pruning and regrafting (SPR). However, these moves are not easy to optimise (Höhna and Drummond 2012; Zhang et al. 2020).

The Billera-Holmes-Vogtmann (BHV) tree space (Billera et al. 2001) is made of one orthant for each topology, connected between themselves at the points where a branch length degenerates to zero to switch from one topology to another. The BHV space is continuous, but its geometry is complex and intrinsically linked with the nature of phylogenetic trees. Consequently, using the BHV is difficult and requires specialised algorithms (Dinh et al. 2017).

Here we propose a new unconstrained Euclidean parametrisation of phylogenetic trees, with fixed number of dimensions, equal to the number of parameters required to specify the branch lengths of a phylogenetic tree. This parametrisation represents topology and branch lengths in a single unconstrained Euclidean space, permitting continuous paths between topologies and providing a foundation for the development of continuous-space computational methods.

The remainder of this article is structured as follows. First we present the general recursive principle used for all constructions, using contour traversal for grafting leaves one by one. Next we show how this can be applied for ultrametric rooted trees, general rooted trees and unrooted trees, respectively. Finally we show numerical validation of the three proposed constructions.

2 Recursive contour-grafting construction

All our parametrisations are constructed recursively using the same basic principle. We consider an arbitrary but fixed order for the leaves. The tree is constructed by adding each leaf one by one to the tree built so far. Let T_i denote the rooted tree formed by the first i leaves and let $L(T_i)$ be the sum of the branch lengths in the tree T_i. The tree T_1 consists only of the first leaf. When adding each leaf i beyond the first one, a graft point x_i is selected on or above T_{i-1} corresponding to the point where the new leaf is attached. This process is repeated for all leaves from the second one ($i = 2$) to the last

one ($i = n$), each time adding to the tree T_{i-1} built previously.

The location of a graft point x_i is parametrised using a variable $a_i \in \mathbb{R}$. If $0 \leq a_i \leq 2L(T_{i-1})$ then a_i corresponds to the distance travelled along the contour traversal of T_{i-1} before reaching the graft point. Starting at the root, the traversal follows each branch once away from the root and once back towards it, visiting the first child of each internal node and then the second. Let $|T|$ denote the set of points on the tree T. The process of finding this graft point on a tree T using the contour traversal is denoted:

$$\gamma_T : [0, 2L(T)] \to |T| \tag{1}$$

We further define the extended contour traversal as illustrated in in Figure 1 and described below. If $a_i > 2L(T_{i-1})$, the grafting point is above the root of T_{i-1} by length $a_i - 2L(T_{i-1})$. Likewise, if $a_i < 0$ then the grafting point is above the root of T_{i-1} by length $-a_i$. This extended contour traversal is therefore a function $\Gamma_T : \mathbb{R} \to |T|^+$. The co-domain $|T|^+$ is obtained by adding to $|T|$ the points on a branch above the root at a distance $d \in [0, \infty)$, with zero identified with the root. The extended contour traversal is defined as:

$$\Gamma_T(a) = \begin{cases} \text{point above root by } -a & \text{if } a < 0 \\ \gamma_T(a) & \text{if } 0 \leq a \leq 2L(T) \\ \text{point above root by } a - 2L(T) & \text{if } a > 2L(T) \end{cases} \tag{2}$$

At the graft point $x_i = \Gamma_{T_{i-1}}(a_i)$, a new node is created with first child equal to what was previously below that point, and second child equal to the new branch of length ℓ_i leading to the leaf i. This convention also defines the ordering of the children used in subsequent contour traversals.

The extended contour traversal is continuous because $\gamma_T(0) = \gamma_T(2(L(T)) = \text{root}(T)$. Every point in the interior of an edge of T is visited exactly twice, once in each traversal direction. A point at distance $d > 0$ above the root also has exactly two preimages, namely $-d$ and $2L(T) + d$. We can deduce the following recursive formula for the sum of the branch lengths $L(T_i)$:

$$L(T_i) = \begin{cases} 0 & \text{if } i = 1 \\ L(T_{i-1}) + \ell_i + \max(0, a_i - 2L(T_{i-1})) + \max(0, -a_i) & \text{otherwise} \end{cases} \tag{3}$$

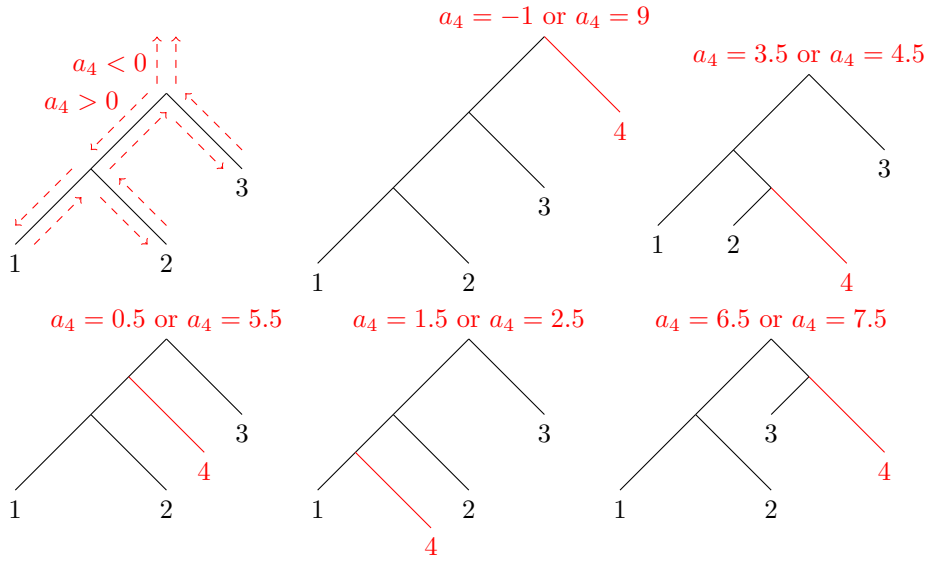


Figure 1: Illustration of the contour traversal. The initial tree is shown in black, with three leaves labeled 1, 2 and 3 and all four branches of length 1. Leaf 4 is added with the value of a_4 determining the graft point as shown and the new terminal branch shown in red.

3 Rooted ultrametric trees

We consider phylogenetic trees that are binary, ultrametric and rooted, with n labeled leaves. There are $n-1$ internal nodes including the root. The topology of the tree is discrete and there are $(2n-3)!! = (2n-2)!/(2^{n-1}(n-1)!)$ unique possibilities (Felsenstein 1978). The branch lengths are continuous but not independent from each other. Since the tree is ultrametric, the branch lengths are constrained so that the number of free parameters corresponds to the $n-1$ values for the heights of the internal nodes relative to the contemporaneous leaves.

The previous section introduced the recursive principle for constructing trees using the extended contour traversal for grafting, with parameters $(a_i)_{i=2,\ldots,n} \in \mathbb{R}^{n-1}$ used in Equation 2. This construction is necessary and sufficient to define an ultrametric rooted tree, since once a graft point has been selected, the newly created terminal branch must stretch from the graft point to the contemporaneous time in which all leaves belong. In particular, the dimensionality of the parameters is $n-1$ which is the dimensionality needed to specify the $n-1$ heights of the internal nodes. For a point x on an ultrametric tree, let $h_T(x)$ denote its distance to any descendant leaf. For a point at

distance d above the current root r, its distance is $h_T(x) = h_T(r) + d$.

Let $\mathcal{C}_n^1 = \mathbb{R}^{n-1}$ denote the parameter space defined above and let $\mathcal{T}_n^1$ denote the space of ultrametric phylogenetic trees with n labelled leaves. An element of $\mathcal{C}_n^1$ is the vector $C = (a_i)_{i=2,\dots,n}$ and it defines a unique phylogenetic tree $T \in \mathcal{T}_n$. This construction defines a function $\Phi_n^1 : \mathcal{C}_n^1 \to \mathcal{T}_n^1$, which is detailed in Algorithm 1. This function is surjective, meaning that for any phylogenetic tree T there is at least one parametrisation C such that $\Phi_n^1(C) = T$. To see this, consider any $T \in \mathcal{T}_n^1$, remove leaf n and its pendant branch, and suppress its resulting degree-two parent. Repeating this operation produces trees with successively fewer leaves. Reversing these steps reconstructs T using the proposed grafting procedure, because every required grafting location is reached by the contour traversal and every terminal branch length $\ell_i > 0$ is deterministic. However, the function Φ_n^1 is not injective as there are more than one C such that $\Phi_n^1(C) = T$. This is a consequence of each grafting point being visited twice by the contour traversal: once as the algorithm goes down the branch and then again as it goes back up the branch. Likewise, a grafting point above the root by length x can be obtained by both $a_i = -x$ and $a_i = 2L(T_{i-1}) + x$. Each grafting point is therefore represented exactly twice if we exclude the possibility of branches of length zero. For an ultrametric rooted binary tree T with strictly positive branch lengths, there are therefore 2^{n-1} distinct values of C such that $\Phi_n^1(C) = T$. The duplication of grafting coordinates is needed to achieve continuity: it allows the grafting location to move continuously through the endpoints of the contour and across changes in tree topology, without an explicit jump between discrete topological states.

Algorithm 1 Parametrisation of ultrametric rooted trees

Require: Ordered leaves $1, \dots, n$ and parameters $C = (a_i)_{i=2,\dots,n} \in \mathbb{R}^{n-1}$
Ensure: An ultrametric rooted phylogenetic tree T_n
Let T_1 consist of the single leaf 1, which is also its root
$L(T_1) \leftarrow 0$
for $i = 2, \dots, n$ **do**
    Find the grafting point $x_i = \Gamma_{T_{i-1}}(a_i)$ using Equation 2
    If x_i lies above the current root, let x_i be the new root
    Insert a new internal node at x_i, splitting its branch if necessary
    Make the subtree previously below x_i the first child of this node
    Attach leaf i as its second child by a branch of length $\ell_i = h_{T_{i-1}}(x_i)$
    Denote the resulting tree by T_i
    $L(T_i) \leftarrow L(T_{i-1}) + \ell_i + \max(0, -a_i) + \max(0, a_i - 2L(T_{i-1}))$
end for
return T_n

To show mathematically that the parametrisation is continuous, we need to consider a tree space with a well-defined associated metric. Let $\mathcal{B}_n^1$ denote the appropriate Billera-Holmes-Vogtmann (BHV) tree space (Billera et al. 2001) of rooted ultrametric phylogenetic trees with n labelled leaves including trees

with zero-length internal branches. The Algorithm 1 can be seen as defining a function $\Phi_n^1 : \mathcal{C}_n^1 \to \mathcal{B}_n^1$ which is continuous. Within each region of parameter space in which the selected contour segments and traversal directions are fixed, all node heights and branch lengths vary continuously with the parameters a_i. When a grafting location passes through an internal node, the separating internal branch contracts continuously to zero before the adjacent topology is entered. Both topologies therefore meet at the same boundary tree in BHV space. The extension above the root also tends continuously to zero when a_i approaches either endpoint of the contour ($a_i = 0$ and $a_i = 2L(T_{i-1})$). Applying this argument recursively at each grafting step establishes the continuity of Φ_n^1.

Finally we calculate the Jacobian of the function $\Phi_n^1 : \mathcal{C}_n^1 \to \mathcal{T}_n^1$. Let $\boldsymbol{h} = (h_2, ..., h_n)$ where h_i is the height of the internal node created when leaf i is grafted. Within a fixed contour region, $h_i = h_{T_{i-1}}(\Gamma_{T_{i-1}}(a_i))$. The new height depends only on $a_2, ..., a_i$, not on later parameters. Moreover,

$$\frac{\partial h_i}{\partial a_i} \in \{-1, +1\}$$

because height changes at unit rate along every fixed contour segment and along the branch above the root. Therefore the Jacobian matrix is triangular, with diagonal entries $+1$ or -1, and

$$\left|\det \frac{\partial(h_2, ..., h_n)}{\partial(a_2, ..., a_n)}\right| = 1$$

Given a density p_T on ultrametric tree space, a natural symmetric pullback assigns equal mass to the 2^{n-1} representations of each generic tree. Since the absolute Jacobian is one, the corresponding density on C is:

$$p_C(C) = \frac{1}{2^{n-1}} p_T(\Phi_n^1(C)) \tag{4}$$

4 Rooted trees with unrestricted branch lengths

We consider phylogenetic trees that are binary and rooted, with n labeled leaves. There are $n-1$ internal nodes including the root. The topology of the tree is discrete and there are $(2n-3)!! = (2n-2)!/(2^{n-1}(n-1)!)$ unique possibilities (Felsenstein 1978). The branch lengths are continuous, and there are $2n-2$ branch lengths taking values in $\mathbb{R}_{>0}$ (one for each leaf and non-root internal node). Our construction is similar to the one above for ultrametric trees. The main difference is that there is now another parameter $b_i \in \mathbb{R}$ for each leaf beyond the first one, that represents the length

$\exp(b_i)$ of the branch separating the graft point (determined by a_i) and the newly added leaf i.

Let $\mathcal{C}_n^2 = \mathbb{R}^{2n-2}$ denote the parameter space defined above and let $\mathcal{T}_n^2$ denote the space of rooted phylogenetic trees with n labelled leaves. An element of $\mathcal{C}_n^2$ is the vector $C = ((a_i, b_i))_{i=2,\dots,n}$ and it defines a unique phylogenetic tree $T \in \mathcal{T}_n^2$. The construction above defines a function $\Phi_n^2 : \mathcal{C}_n^2 \to \mathcal{T}_n^2$, which is detailed in Algorithm 2. Following the same arguments as outlined above for ultrametric trees, we can show that this function is continuous and surjective. It is not injective, instead each rooted binary tree with strictly positive branch lengths has exactly 2^{n-1} preimages.

Algorithm 2 Parametrisation of rooted phylogenetic trees

Require: Ordered leaves $1, \dots, n$ and parameters $C = ((a_i, b_i))_{i=2,\dots,n} \in \mathbb{R}^{2n-2}$
Ensure: A rooted phylogenetic tree T_n
  Let T_1 consist of the single leaf 1, which is also its root
  $L(T_1) \leftarrow 0$
  for $i = 2, \dots, n$ **do**
    Find the grafting point $x_i = \Gamma_{T_{i-1}}(a_i)$ using Equation 2
    If x_i lies above the current root, let x_i be the new root
    Insert a new internal node at x_i, splitting its branch if necessary
    Make the subtree previously below x_i the first child of this node
    Attach leaf i as its second child by a branch of length $\exp(b_i)$
    Denote the resulting tree by T_i
    $L(T_i) \leftarrow L(T_{i-1}) + \exp(b_i) + \max(0, -a_i) + \max(0, a_i - 2L(T_{i-1}))$
  end for
  return T_n

As previously, we calculate the Jacobian of the function $\Phi_n^2 : \mathcal{C}_n^2 \to \mathcal{T}_n^2$. We consider this Jacobian locally, within a region of $\mathcal{C}_n$ where the resulting topology and the branch containing each grafting point are fixed. Let $\boldsymbol{\ell}^{(i-1)}$ denote the vector of branch lengths in T_{i-1}. If leaf i is grafted in the interior of a branch of length ℓ_j, that branch is replaced by two branches of lengths

$$u_i(\boldsymbol{\ell}^{(i-1)}, a_i) \quad \text{and} \quad \ell_j - u_i(\boldsymbol{\ell}^{(i-1)}, a_i)$$

and a terminal branch of length $\exp(b_i)$ is added. Along any fixed segment and direction of the contour traversal, u_i is affine in a_i, with

$$\frac{\partial u_i}{\partial a_i} \in \{-1, 1\}$$

After reordering the branch-length coordinates, the absolute Jacobian determinant of this grafting step is therefore

$$\left| \det \frac{\partial \boldsymbol{\ell}^{(i)}}{\partial (\boldsymbol{\ell}^{(i-1)}, a_i, b_i)} \right| = \left| \frac{\partial u_i}{\partial a_i} \right| \frac{\partial \exp(b_i)}{\partial b_i} = \exp(b_i)$$

The same result holds when the grafting point lies above the root, since the new root branch has derivative -1 or $+1$ with respect to a_i. Applying recursively gives, for every C in which the grafting

points do not coincide with a node or an endpoint of the contour:

$$\left|\det \frac{\partial \boldsymbol{\ell}(\Phi_n(C))}{\partial C}\right| = \prod_{i=2}^{n} \exp(b_i) = \exp\left(\sum_{i=2}^{n} b_i\right)$$

Given a density p_T on rooted tree space, a natural symmetric pullback assigns equal mass to the 2^{n-1} representations of each generic tree. The corresponding density on C is therefore:

$$p_C(C) = p_T(\Phi_n^2(C)) \frac{1}{2^{n-1}} \exp\left(\sum_{i=2}^{n} b_i\right) \tag{5}$$

5 Unrooted trees

TODO using the unextended contour traversal in Equation 1.

6 Numerical validation

We consider rooted trees, such that all of the $(2n-3)!!$ topologies are equally likely and the branch lengths ℓ_i follow an Exponential distribution with mean θ, so that:

$$p_T(T) = \frac{1}{(2n-3)!!} \prod_{i=1}^{2n-2} \theta^{-1} \exp\left(-\frac{\ell_i}{\theta}\right) \tag{6}$$

To sample from this distribution, we use the adaptive Metropolis sampler (Vihola 2011). This algorithm uses a multivariate normal distribution as the proposal kernel $q(C'|C)$ which is symmetric ($q(C'|C) = q(C|C')$), so that the acceptance probability is simply $\min(1, p_C(C')/p_C(C))$ calculated using Equation 5. The mean of the proposal kernel is the current value C and its covariance matrix is tuned to achieve an acceptance rate of 0.234 (Andrieu and Thoms 2008). We take $n = 5$ as a small example, with $\theta = 2$ as the mean of the distribution of branch lengths. The sampler was run for 2×10^5 iterations, with the first half discarded as burn-in and the remainder sampled every 100 iterations to produce a sample of size 1000. This run took less than a minute on a standard laptop. Figure 2 shows that the trees sampled by the MCMC are distributed as expected under Equation 6. All $(2n-3)!! = 105$ topologies have been sampled, the distribution of the sum of branch lengths and of

a single branch length are as expected, and a majority-rule consensus tree looks completely star-like. The effective sample size was greater than 200 for individual branch lengths as computed using the R package coda (Plummer et al. 2006). The approximate effective sample size of the tree topology was also greater than 200 as computed by the R package rwty (Warren et al. 2017).

7 Discussion

The aim of Bayesian phylogenetic inference is to reconstruct a tree given some genetic data using Bayesian statistics to capture the posterior distribution of the tree (Nascimento et al. 2017). This is usually done using a Markov Chain Monte-Carlo (MCMC) to generate samples from this posterior distribution (Gilks et al. 1996). The MCMC needs to propose updates to both the branch lengths and the topology, with the latter using topology-changing moves such as nearest-neighbor interchange (NNI) or subtree pruning and regrafting (SPR). However, these moves are not easy to optimise (Höhna and Drummond 2012; Zhang et al. 2020). Conversely, many promising MCMC methods have not been applied in Bayesian phylogenetics because they can only be applied to continuous parameters. This includes in particular some adaptive methods where proposal parameters are learned during sampling (Andrieu and Thoms 2008), as well as gradient-based approaches such as the Langevin and Hamiltonian methods (Roberts and Rosenthal 1998; Neal 2011; Betancourt 2017). For example the dynamic Hamiltonian Monte Carlo (HMC) method implemented in Stan (Carpenter et al. 2017; Betancourt 2017). After observing data D made of n aligned genetic sequences of length l, the distribution becomes the posterior with density $p_C(C|D) \propto p_C(C)p(D|T = \Phi_n(C))$where the last term is the likelihood of the data given the phylogenetic tree $T = \Phi_n(C)$ obtained using Algorithm 2. This likelihood can be calculated efficiently using the Felsenstein pruning algorithm (Felsenstein 1981) which is used in most phylogenetic inference tools. Many substitution models can be used to calculate this likelihood (Nascimento et al. 2017) and here we use the simplest Jukes-Cantor model (Jukes and Cantor 1969) which avoids having to infer additional parameters.

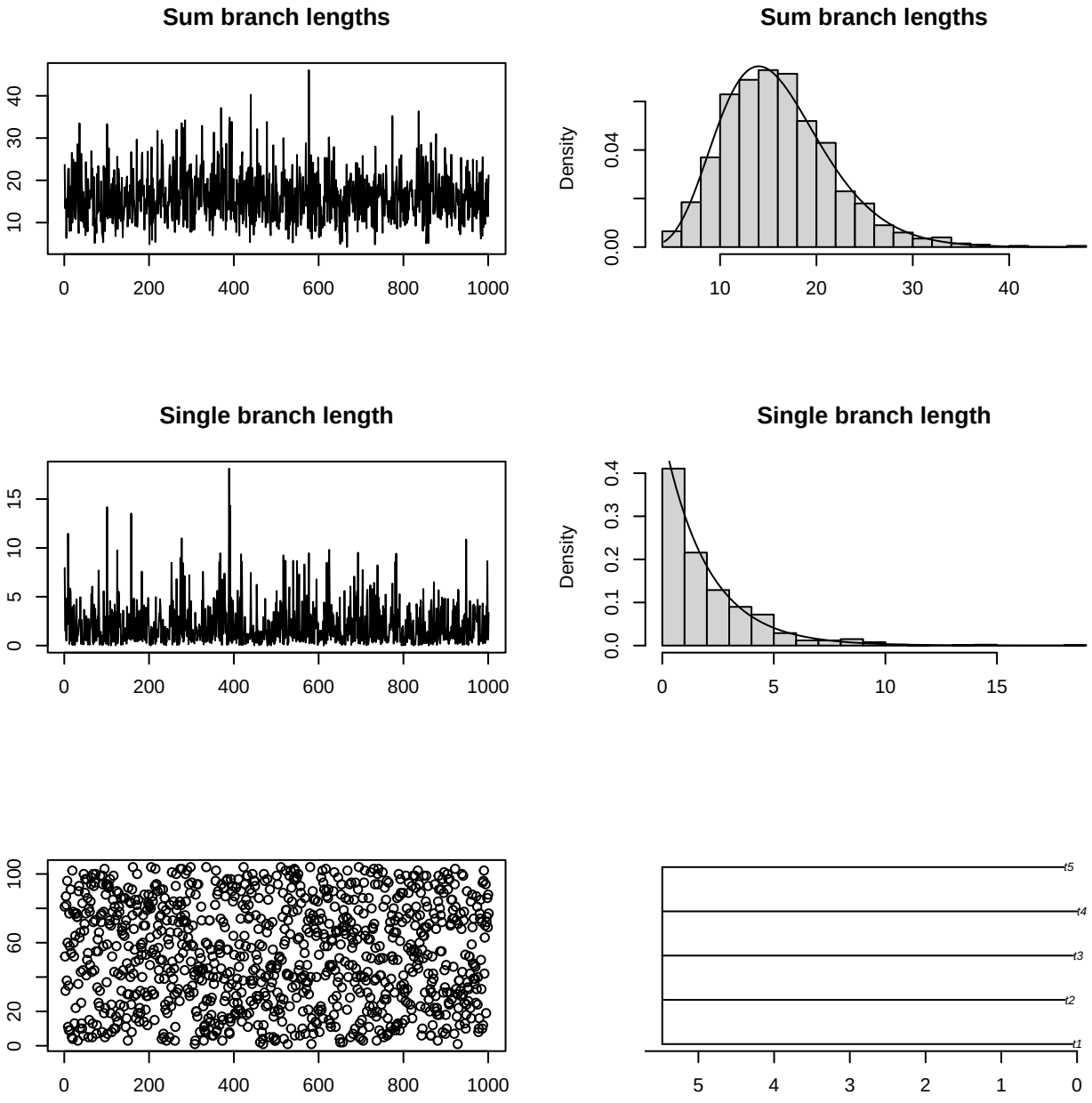


Figure 2: MCMC run. (A) Trace of sum of branch lengths. (B) Distribution of sum of branch lengths, sampled (histogram) and expected (line). (C) Trace of a single branch length. (D) Distribution of a single branch length, sampled (histogram) and expected (line). (E) Trace of topology. (F) Majority-rule consensus tree.

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